

Part 1:

Edge Impulse Link: <https://studio.edgeimpulse.com/public/995775/live>

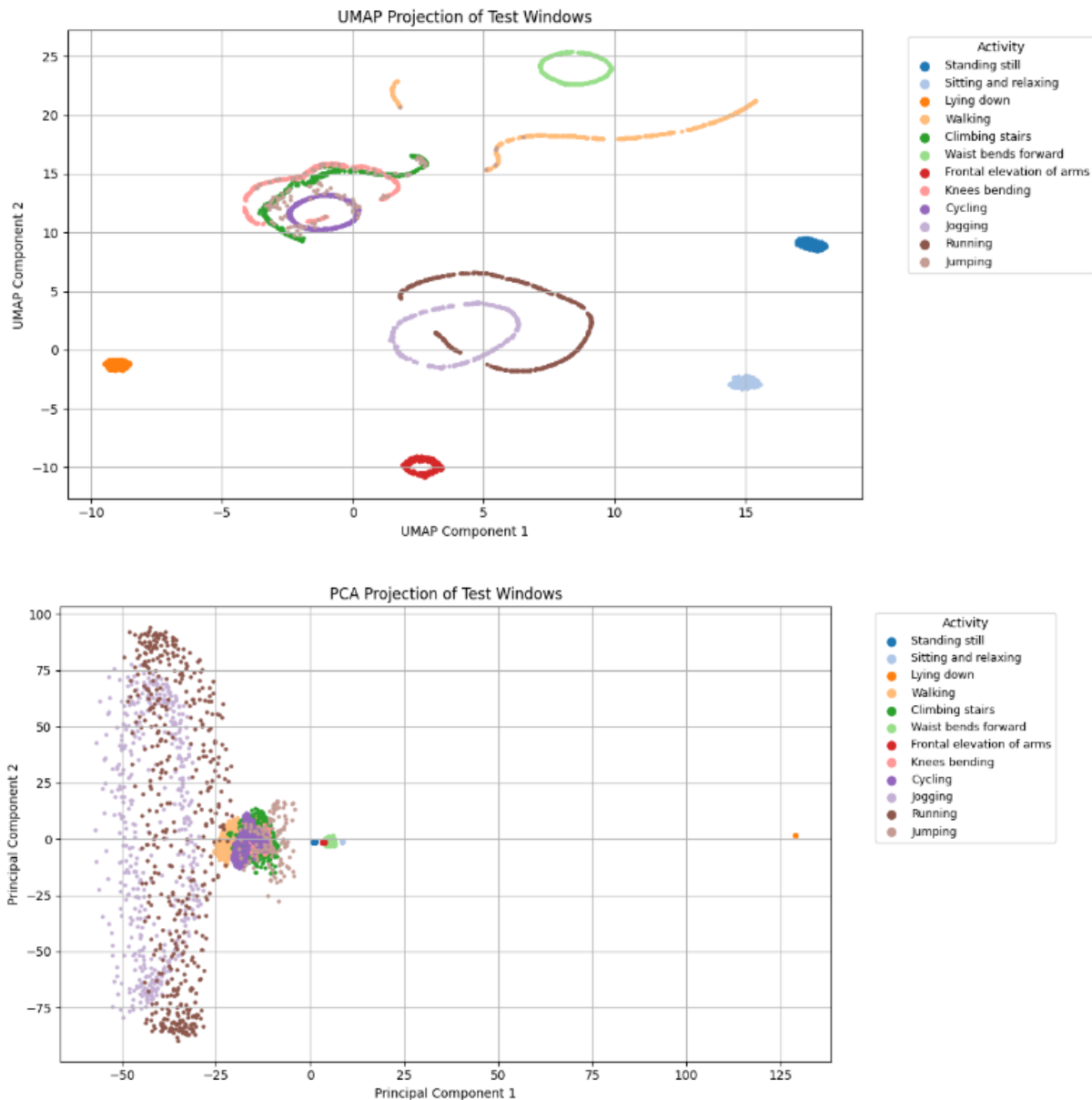
1. Raw inputs were vibration sensor signals collected from the fan system. The data is a time series of accelerometer measurements
2. The types of features are spectral and statistical, extracted from the accelerometer signals. There is a total of 33 input features that were passed into the neural network classifier, which include: accX RMS, accX Peak 1 Height, accY Peak 3 Height, accX Peak 1 Freq, accY RMS
3. The outputs of the NN classifier are: “nominal” and “off”. It produces scores for each class, indicating how likely the current sensor data is to belong to either a normal operating condition or an anomalous condition.
4. The anomaly detector used selected vibration features extracted from accelerometer data. Four features were used for anomaly detection training: accX RMS, accX Peak 1 Height, accY Spectral Power 2.0-5.0 Hz, accZ Spectral Power 2.0-5.0 Hz.

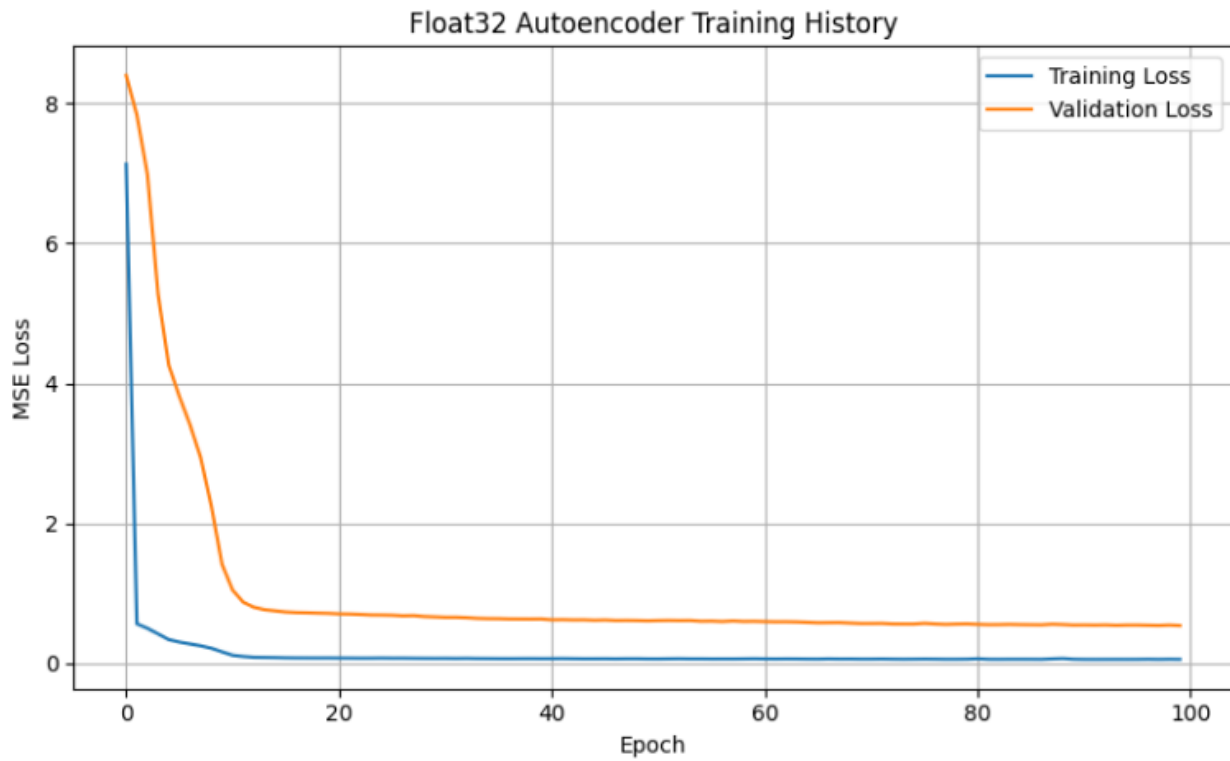
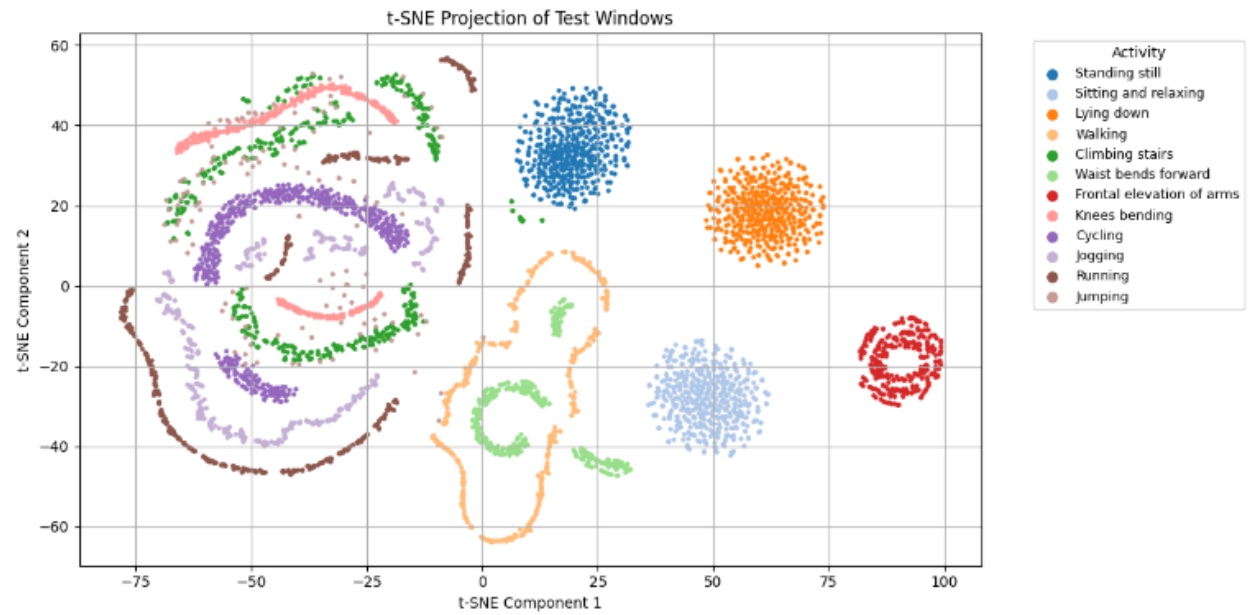
```
Anomaly prediction: 16.023212
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 49 ms, inference 613 us, anomaly 505 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.024956
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 45 ms, inference 595 us, anomaly 516 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.843750
  off: 0.156250
Anomaly prediction: 477.814453
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 49 ms, inference 593 us, anomaly 525 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.785156
  off: 0.214844
Anomaly prediction: 331.171356
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 45 ms, inference 609 us, anomaly 525 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.023437
  off: 0.976562
Anomaly prediction: 13.860880
Starting inferencing in 2 seconds...
```

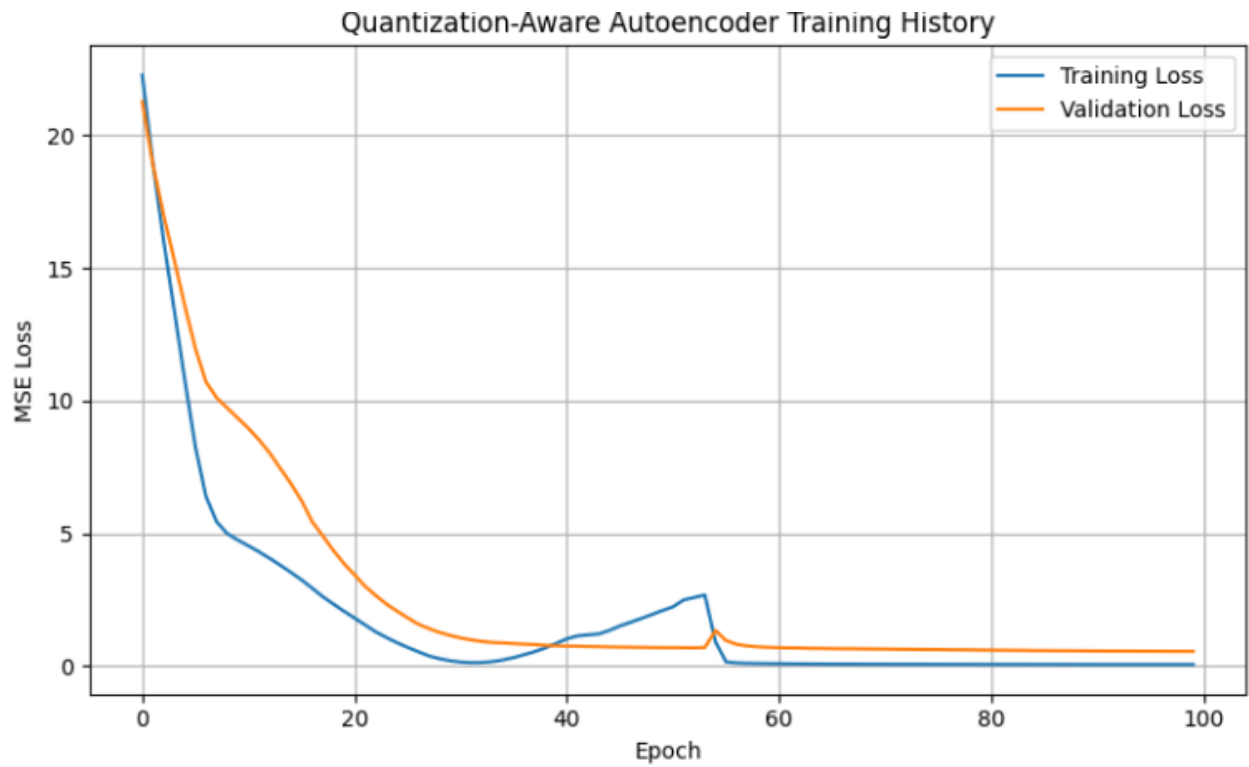
- 5.
6. When deploying, it produced low anomaly scores when the board remained stationary. However, slowly moving the board left-right and up-down caused the anomaly score to increase significantly, even when the classifier still produced relatively high nominal probabilities. This shows that the classifier's output should not always be fully trusted, because the model may still assign high confidence to a

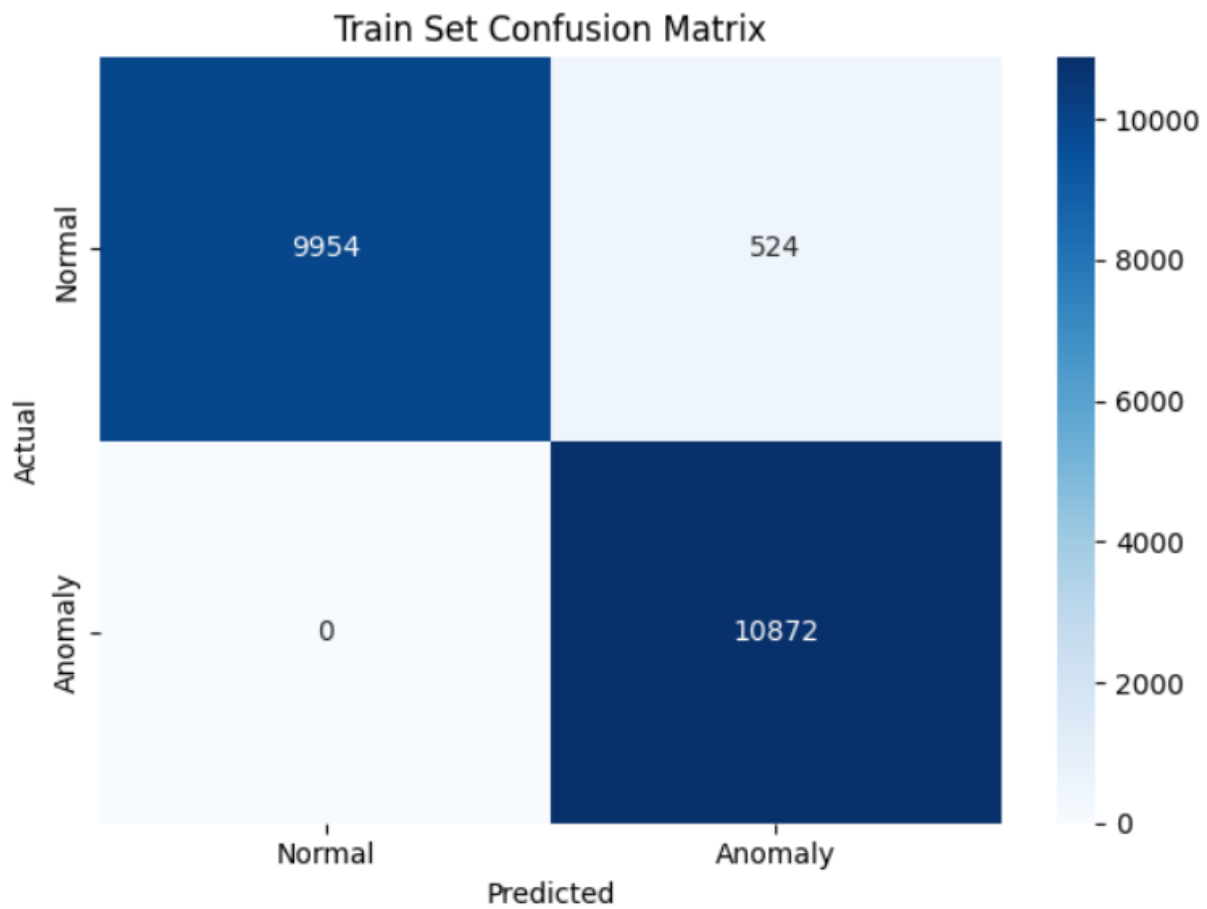
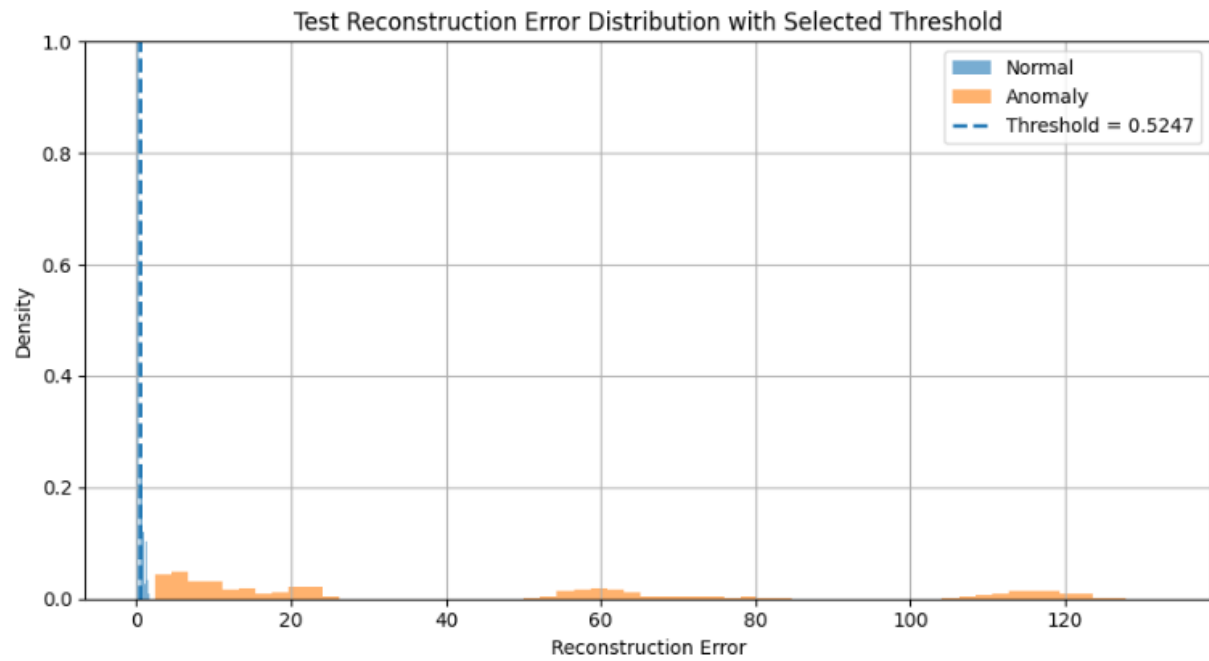
known class even when the signal differs from the original training distribution. The anomaly detector helped identify unusual or unseen vibration patterns that the classifier alone did not fully capture.

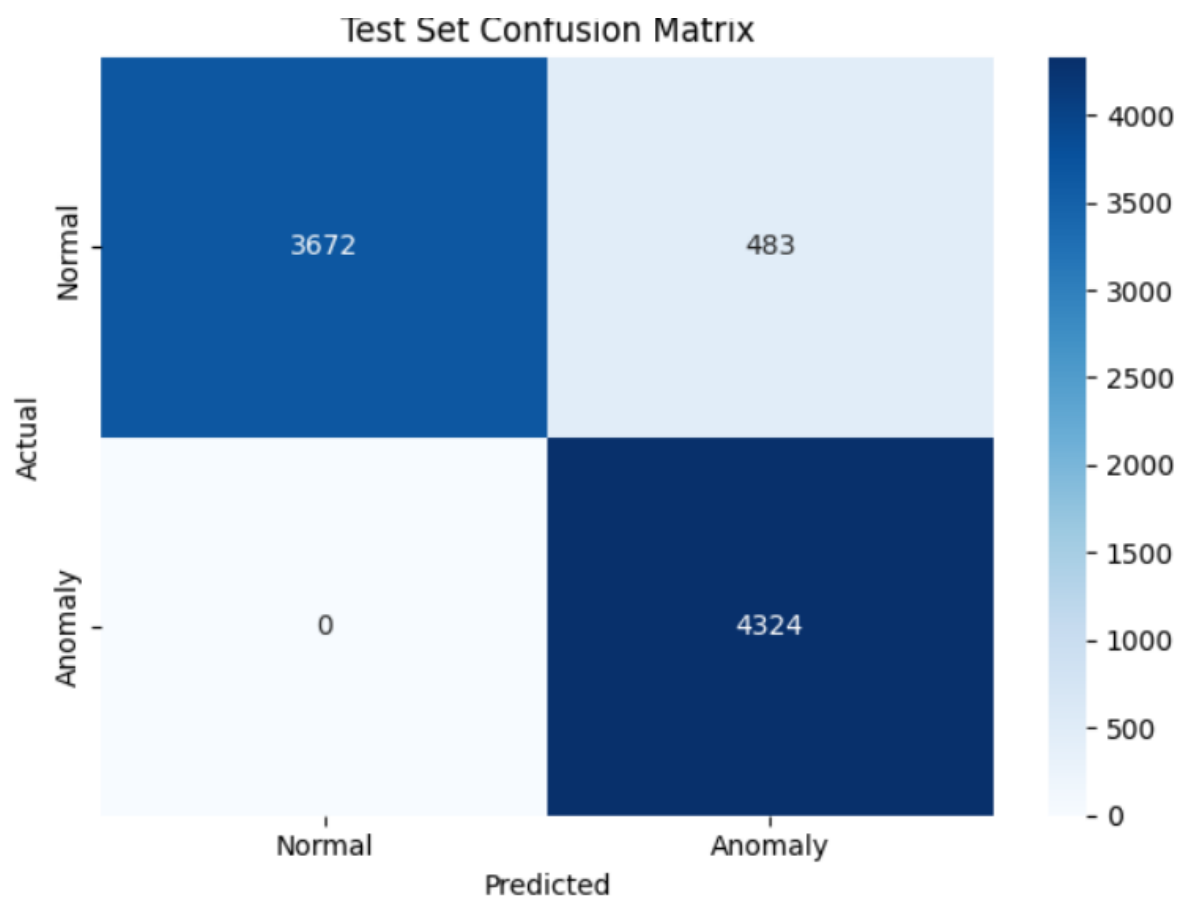
Part 2:

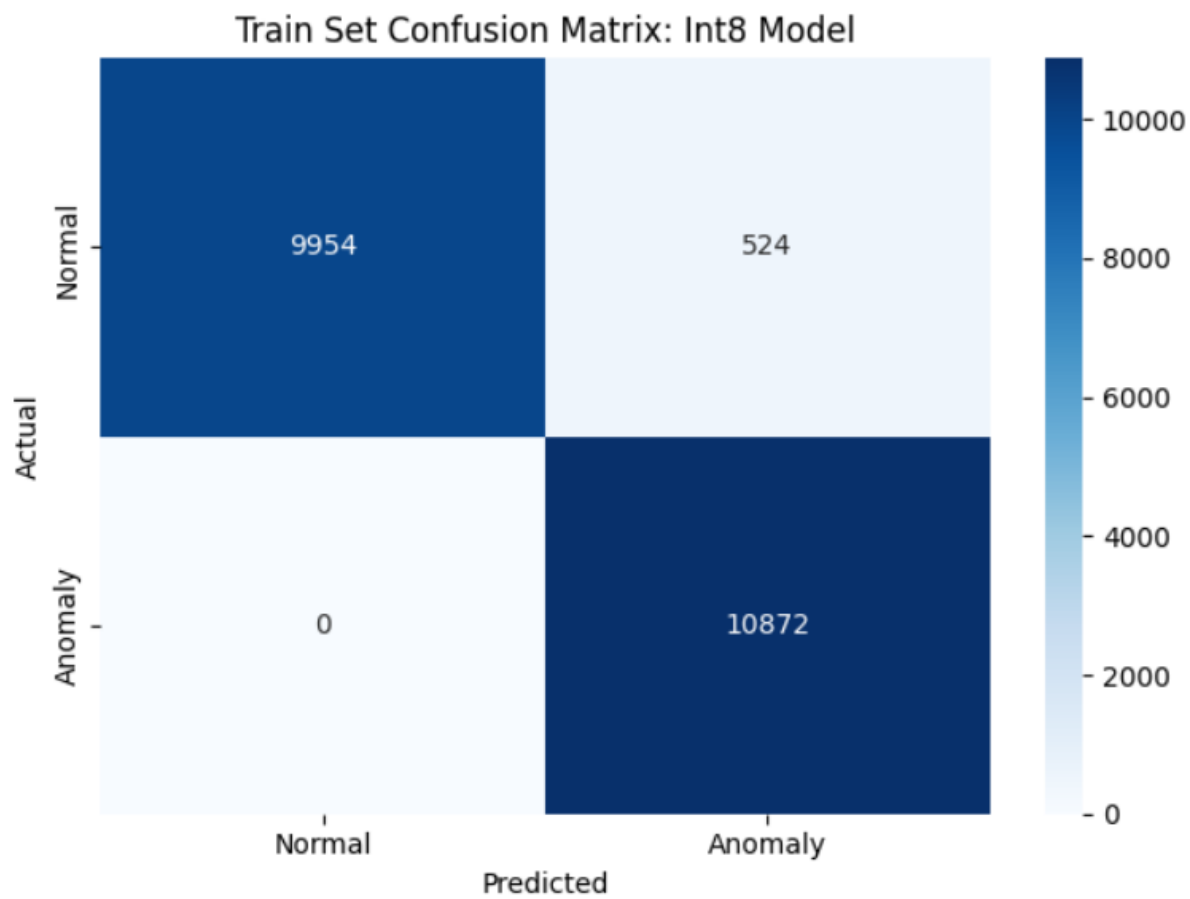


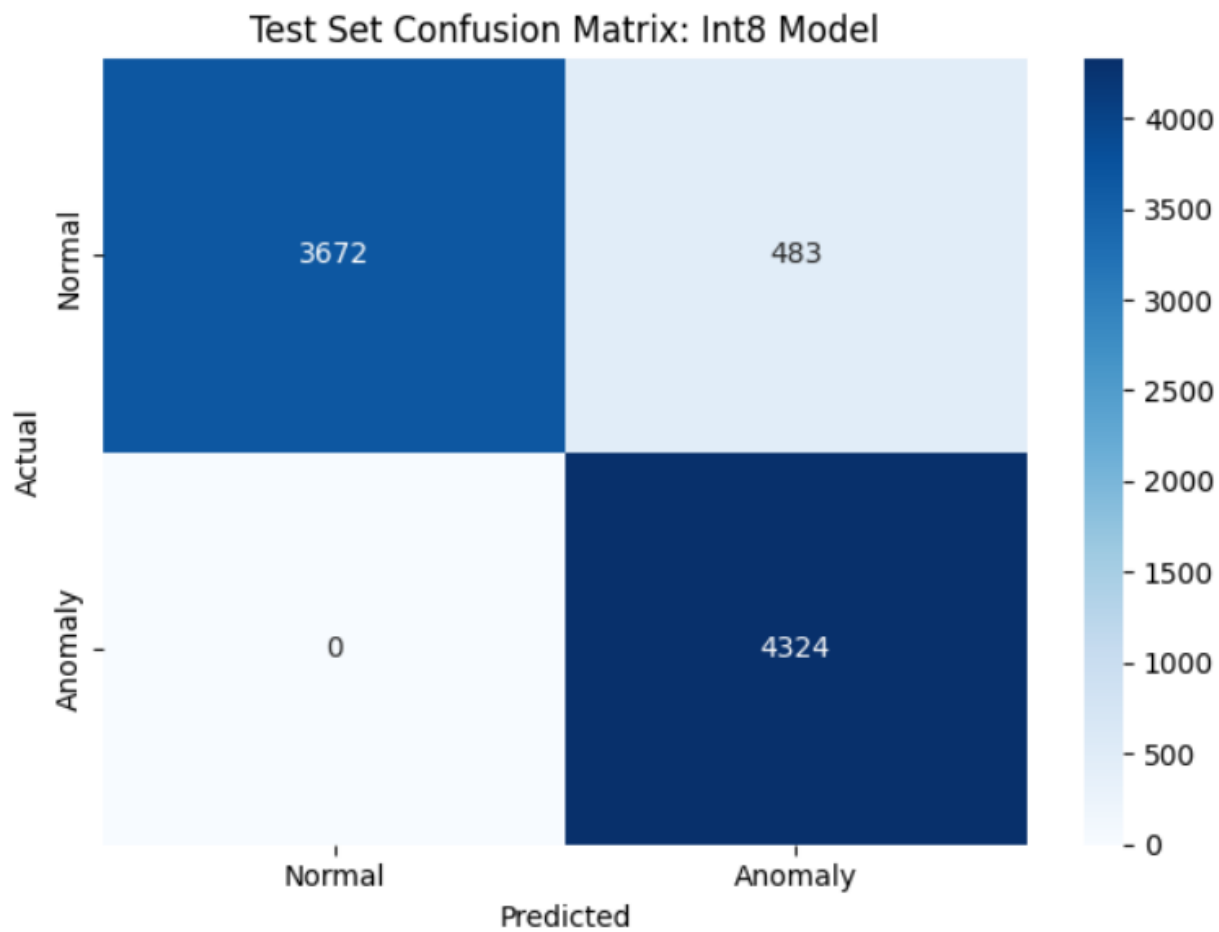












Train set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy			0.98	21350
macro avg	0.98	0.97	0.98	21350
weighted avg	0.98	0.98	0.98	21350

Test set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	0.88	0.94	4155
Anomaly (1)	0.90	1.00	0.95	4324
accuracy			0.94	8479
macro avg	0.95	0.94	0.94	8479
weighted avg	0.95	0.94	0.94	8479

Train set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	0.95	0.97	10478
Anomaly (1)	0.95	1.00	0.98	10872
accuracy			0.98	21350
macro avg	0.98	0.97	0.98	21350
weighted avg	0.98	0.98	0.98	21350

Test set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	0.88	0.94	4155
Anomaly (1)	0.90	1.00	0.95	4324
accuracy			0.94	8479
macro avg	0.95	0.94	0.94	8479
weighted avg	0.95	0.94	0.94	8479

```
Reconstruction error: 0.753536
Result: Normal activity
Reconstruction error: 0.846399
Result: Normal activity
Reconstruction error: 0.904447
Result: Normal activity
Reconstruction error: 1.097229
Result: Normal activity
Reconstruction error: 1.166849
Result: Normal activity
Reconstruction error: 1.182582
Result: Normal activity
Reconstruction error: 1.251101
Result: Normal activity
Reconstruction error: 1.310909
Result: Normal activity
Reconstruction error: 1.357763
Result: Normal activity
Reconstruction error: 1.493427
Result: Normal activity
Reconstruction error: 1.545705
Result: Anomaly detected
Reconstruction error: 1.610774
Result: Anomaly detected
Reconstruction error: 1.728234
Result: Anomaly detected
Reconstruction error: 1.828179
Result: Anomaly detected
```

```
Reconstruction error: 1.988345  
Result: Anomaly detected  
Reconstruction error: 1.918553  
Result: Anomaly detected  
Reconstruction error: 1.887271  
Result: Anomaly detected  
Reconstruction error: 1.823336  
Result: Anomaly detected  
Reconstruction error: 1.757436  
Result: Anomaly detected  
Reconstruction error: 1.712054  
Result: Anomaly detected  
Reconstruction error: 1.719696  
Result: Anomaly detected  
Reconstruction error: 1.628226  
Result: Anomaly detected  
Reconstruction error: 1.620943  
Result: Anomaly detected  
Reconstruction error: 1.596369  
Result: Anomaly detected  
Reconstruction error: 1.502033  
Result: Anomaly detected  
Reconstruction error: 1.442452  
Result: Normal activity  
Reconstruction error: 1.428403  
Result: Normal activity  
Reconstruction error: 1.361449  
Result: Normal activity
```

1. The threshold was selected using the 95th percentile of the normal training reconstruction errors. This allowed most normal activities to remain below the threshold while anomalous activities produced significantly larger reconstruction errors.

2. Reconstruction error measures how well the autoencoder recreates the input data. Normal activities produce low reconstruction error because the model was trained on them, while anomalous activities produce higher reconstruction error since they differ from the learned patterns.
3. The deployed Arduino model must preserve the trained int8 model, reconstruction error threshold, input shape, and preprocessing steps used in the Python notebook so that inference on the microcontroller matches the original training pipeline.
4. It must use the same preprocessing steps as the notebook because the model was trained using that exact data format and feature structure. Different preprocessing could change the input distribution and lead to incorrect reconstruction errors and unreliable anomaly detection results.
5. The main limitations include limited memory and computation resources on the microcontroller, reduced model precision after int8 quantization, and sensitivity to unseen sensor conditions or motion patterns. Changes in sensor placement, noise, or preprocessing can also reduce the reliability of anomaly detection.